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Statistical Methods to Reduce Bias in Binary Endpoints in Dose-finding Studies

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Introduction & Previous Work

- Optimization of dose finding and selection is a key goal in oncology drug research.
- OPTIMUS Project: FDA initiative to improve dose optimization/dose finding.
- Key challenge: Accurate estimation of therapeutic effects
 - True treatment effect is often overestimated in Phase II
 - Phase III results may not confirm Phase II findings.
- Bias-correction methods were presented for normally distributed outcomes (e.g. bootstrap methods) [Zhan2024]
- No existing methods for binary endpoints identified in a literature review.

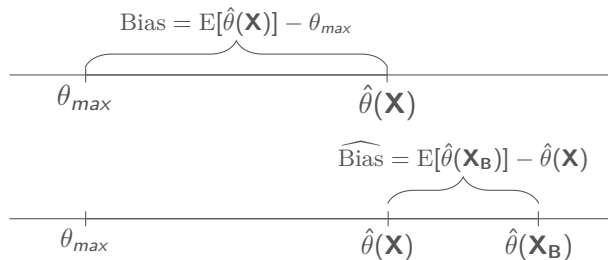
Setup

- l active treatment arms with n patients per arm.
- Response X_{ij} for patient j in group i with $X_{ij} \sim \text{Ber}(\theta_i)$ and $i = 1, \dots, l$.
- $\mathbf{X}_i = (X_{i1}, \dots, X_{in})$ vector of responses for group i and $\mathbf{X} = (\mathbf{X}_1, \dots, \mathbf{X}_l)$.
- Parameter of interest is $\theta_{\max} = \max(\theta_1, \dots, \theta_l)$.
- Estimator: $\hat{\theta}(\mathbf{X}) = \max[\bar{\mathbf{X}}_1, \dots, \bar{\mathbf{X}}_l]$.
- Even if $\bar{\mathbf{X}}_i = \text{mean}(X_{i1}, \dots, X_{in})$ is an unbiased estimator, $\hat{\theta}(\mathbf{X})$ is biased.

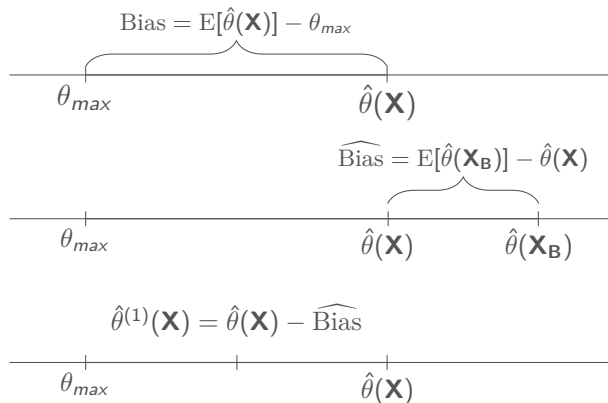
Idea of the Bootstrap Method

$$\text{Bias} = E[\hat{\theta}(\mathbf{X})] - \theta_{max}$$


Idea of the Bootstrap Method



Idea of the Bootstrap Method



Part I - Optimization of Bootstrap Repetitions

Aim: Optimize the number of bootstrap repetitions B for the single and double (non-)parametric bootstrap in binary endpoints, based on bias and MSE.

- The number of bootstrap repetitions B is set to 80, 100, 200, 500, and 1000.

Part II - Bayesian Hierarchical Models

Aim: Identify the optimal hyperprior (with corresponding parameters) for a Bayesian hierarchical model to minimize bias and MSE.

The model is specified as: $Y_i \sim \text{Bin}(n, \theta_i)$, $i = 1, \dots, I$ with

$$\text{logit}(\theta_i) \sim \mathcal{N}(\mu, \sigma^2), \quad \mu \sim \mathcal{N}(-1.38, 100) \text{ and } \sigma^2 \sim F.$$

- Hyperpriors determine how much information is pooled across groups (i.e., σ^2 controls degree of shrinkage).
- The following MCMC settings are applied for all fits: number of chains 3, total iterations per chain: 10 000, burn-in: 2 000.

Part II - Priors

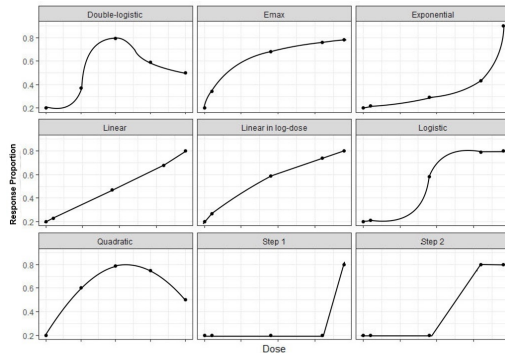
The priors and corresponding parameters to evaluate are based on [Cunanan et al.2019]:

- $\text{Unif}[0, a]$ for $\log(\sigma)$, σ , σ^2 with
 - $a \in \{1, \dots, 1000\}$.
- Half-t for σ with
 - scale parameter $s \in \{0.5, \dots, 1000\}$ and
 - degrees of freedom $df \in \{1, \dots, 500\}$.
- Gamma distribution $\Gamma(a, b)$ for $\tau = \frac{1}{\sigma^2}$ with
 - $a \in \{0.005, \dots, 2\}$, and
 - $b \in \{0.0005, \dots, 50\}$.

General Setup

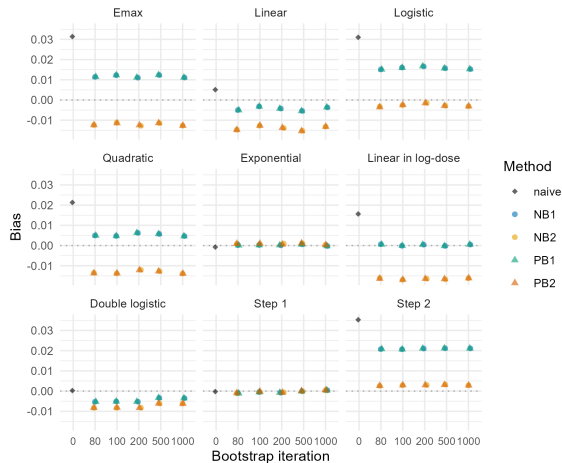
The simulation study consists of two parts (10.000 repetitions each), with the following settings:

- 9 dose response relationships; 3 or **5** doses [Ivanova et al.2012]
- Maximal response: 60% or **80%**
- Sample size: $n = 20$ or **40** patients per dose level

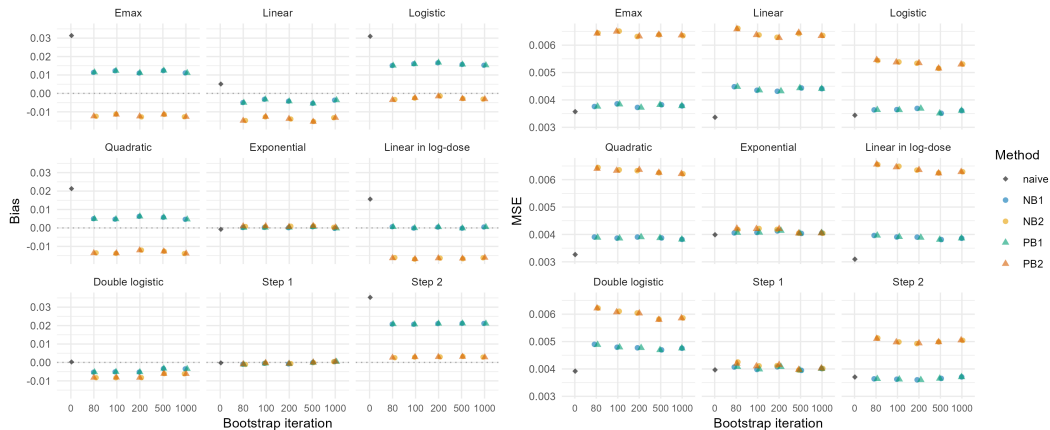


Part I

Part I



Part I

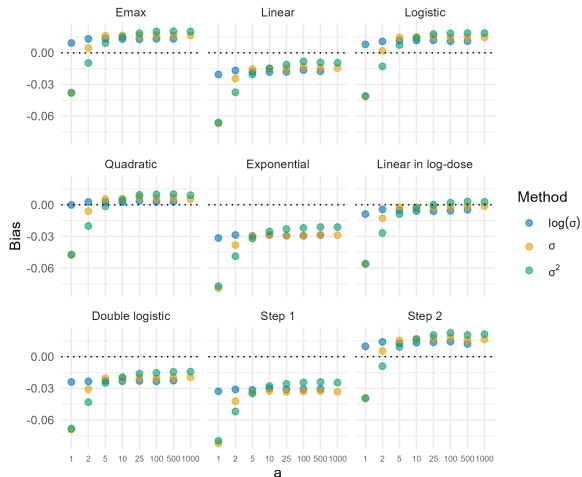


Part I: Summary

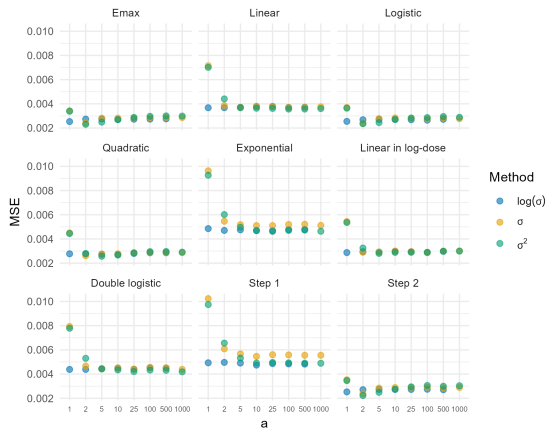
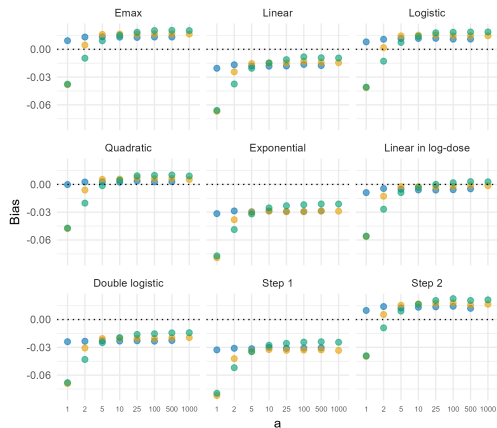
- No relevant differences in bias or MSE across bootstrap iterations.
- $B = 80$ bootstrap iterations provide adequate bias correction.
- Other simulation scenarios reached the same conclusion.
- Performance of the bootstrap method depends on the true dose-response relationship.

Part II: Uniform $[0, a]$

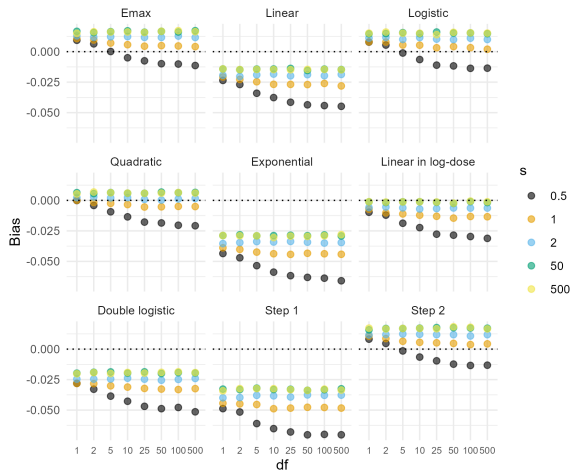
Part II: Uniform [0, a]



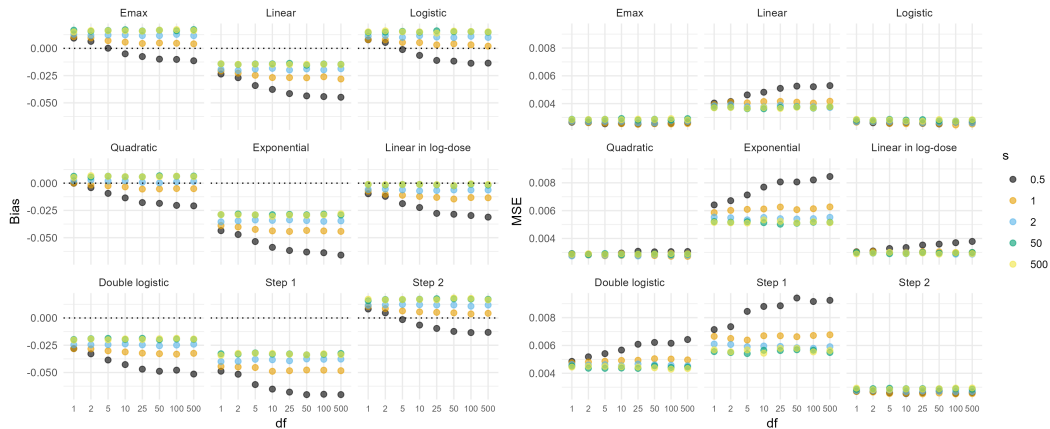
Part II: Uniform $[0, a]$



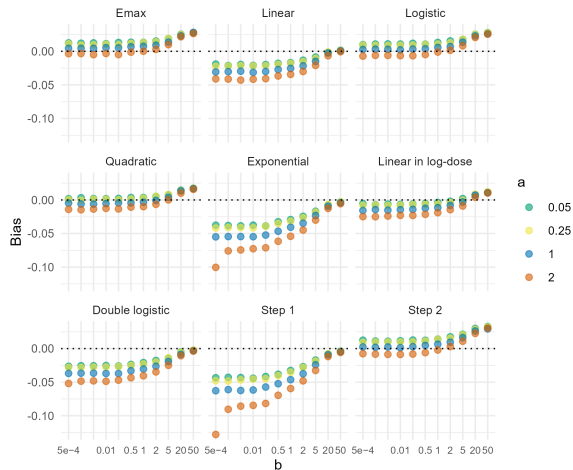
Part II: Half-t(df, s)



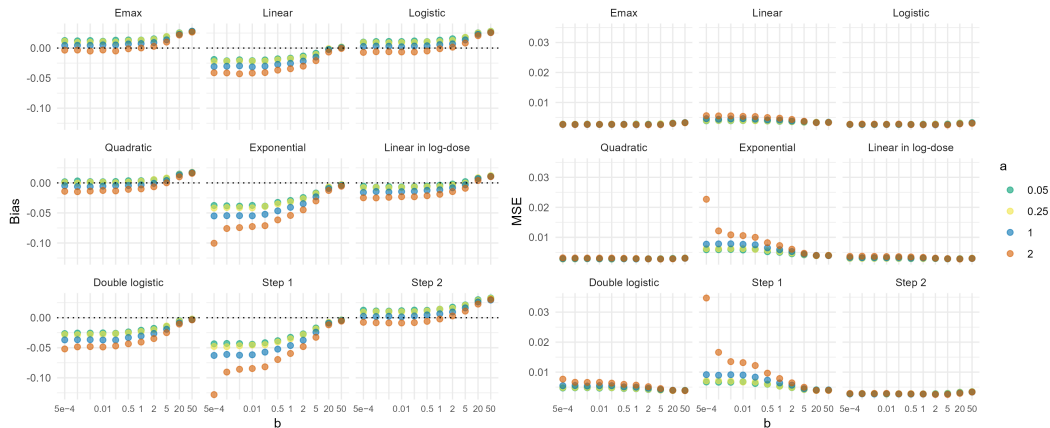
Part II: Half-t(df, s)



Part II: Gamma(a,b)



Part II: Gamma(a,b)



Part II: Summary

Optimal parameters based on mean bias (MSE) across all dose-response relationships:

Method	Parameter				Bias (MSE)
Uniform $\log(\sigma)$	$a =$	2			-0.0070 (0.00351)
Uniform σ	$a =$	100			-0.0047 (0.00371)
Uniform σ^2	$a =$	1000			0.0003 (0.00357)
Half-t	$df =$	100	$s =$	500	-0.0043 (0.00369)
Gamma	$a =$	0.25	$b =$	5	0.0004 (0.00335)

- Uniform prior on σ^2 with $a = 1000$ and Gamma prior with $a = 0.25$ and $b = 5$ show the smallest bias.
- Prior choice depends on the true underlying dose-response scenario.

Conclusion

- $B = 80$ bootstrap iterations are sufficient.
- Bayesian methods show promising results in reducing bias.
- Results depend on the true dose-response relationship.
- Further scenarios for Bayesian methods ($n = 20$, $I = 3$, etc.) and further methods (e.g. Jackknife) will be investigated.

References I

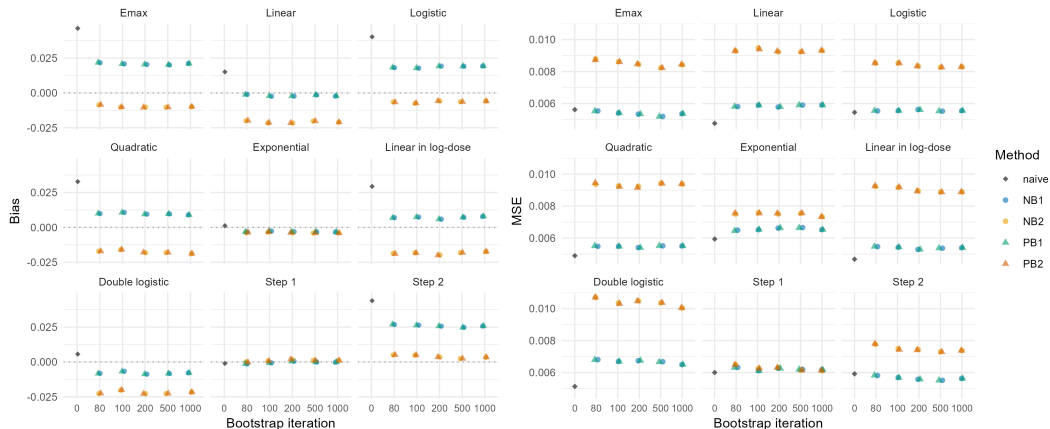
Kristen M Cunanan, Alexia Iasonos, Ronglai Shen, and Mithat Gönen. Variance prior specification for a basket trial design using bayesian hierarchical modeling. *Clinical Trials*, 16(2):142–153, 2019. doi: 10.1177/1740774518812779. URL <https://doi.org/10.1177/1740774518812779>. PMID: 30526008.

Anastasia Ivanova, Changfu Xiao, and Yevgen Tymofyeyev. Two-stage designs for phase 2 dose-finding trials. *Statistics in Medicine*, 31(24):2872–2881, 2012. doi: <https://doi.org/10.1002/sim.5365>. URL <https://onlinelibrary.wiley.com/doi/abs/10.1002/sim.5365>.

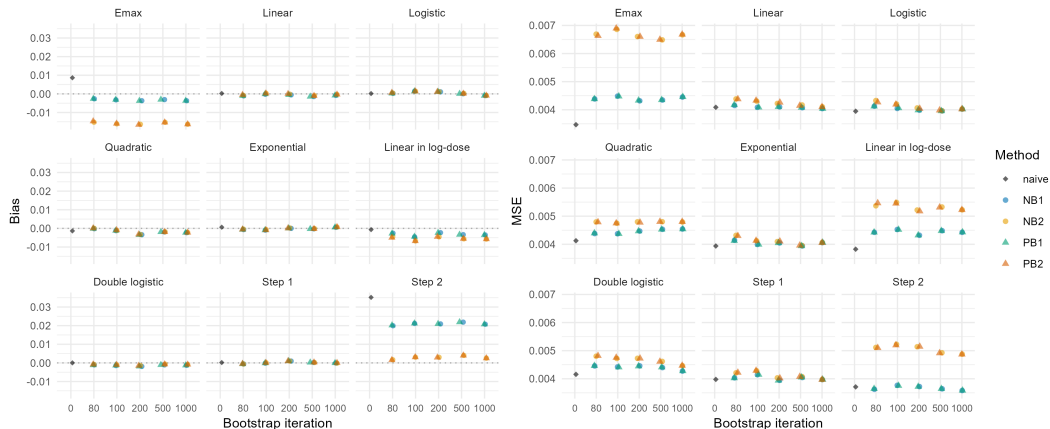
References II

Tianyu Zhan. A class of computational methods to reduce selection bias when designing phase 3 clinical trials. *Statistics in Medicine*, 43(10):1993–2006, 2024. doi: <https://doi.org/10.1002/sim.10041>. URL <https://onlinelibrary.wiley.com/doi/abs/10.1002/sim.10041>.

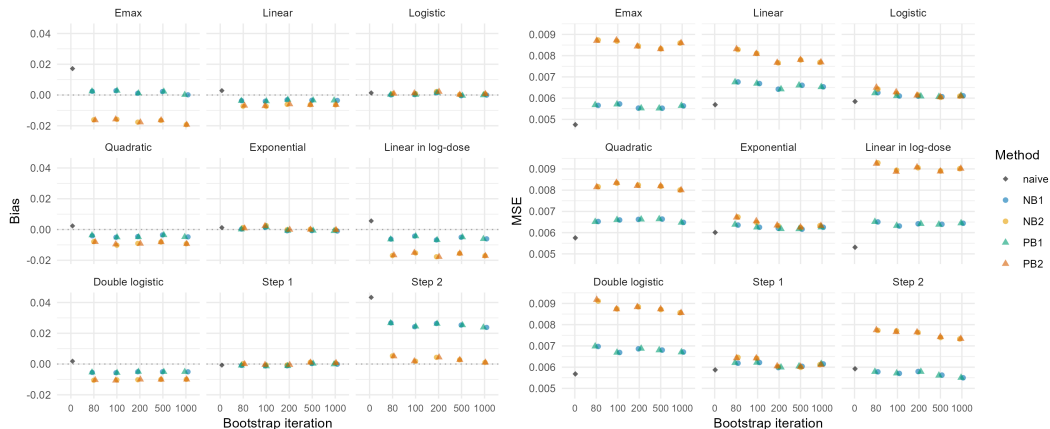
n = 40, max response 60%, 5 groups



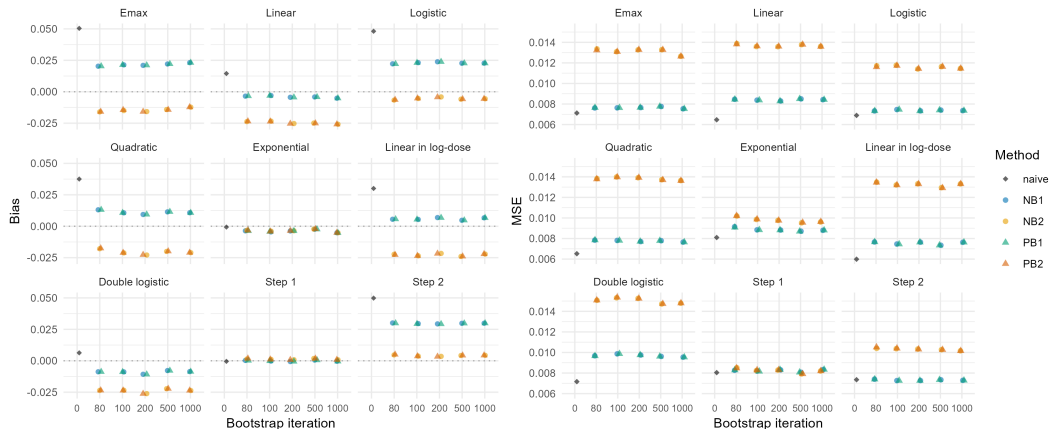
n = 40, max response 80%, 3 groups



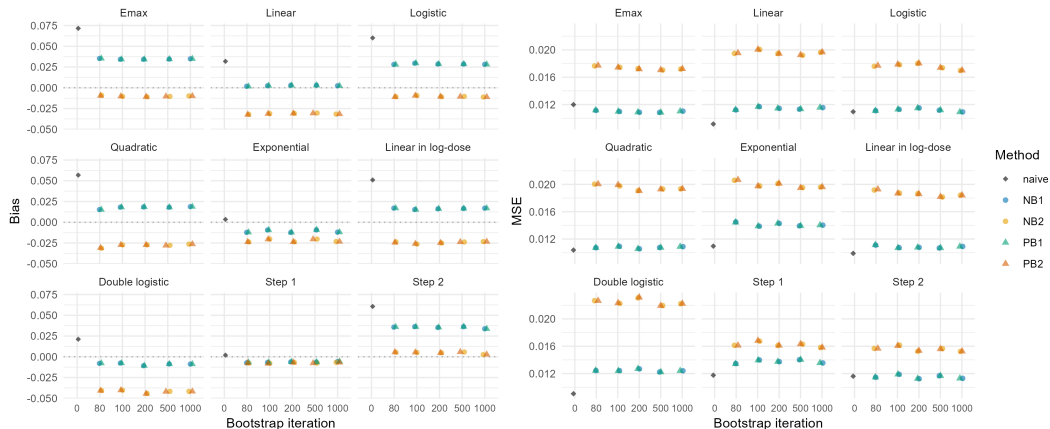
n = 40, max response 60%, 3 groups



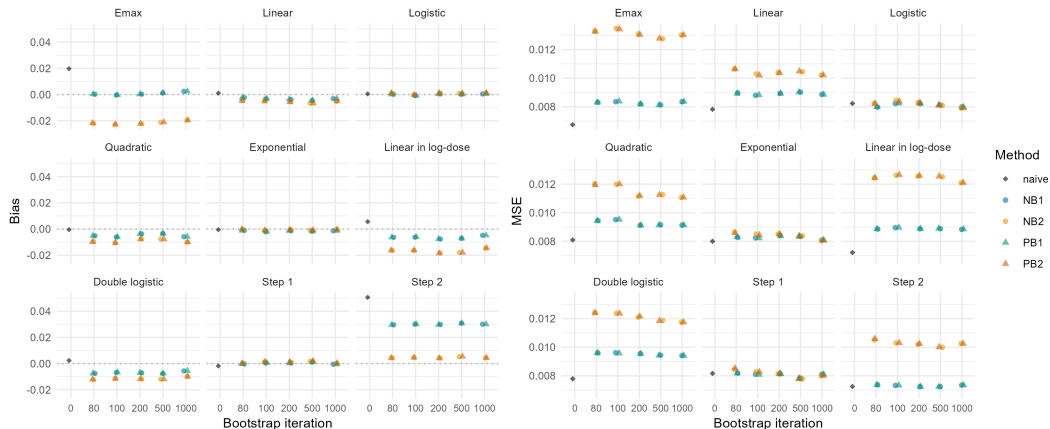
n = 20, max response 80%, 5 groups



n = 20, max response 60%, 5 groups



n = 20, max response 80%, 3 groups



n = 20, max response 60%, 3 groups

